

Data management is people management: on abstraction of data and labour in archaeological projects

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Abstract

Archaeological research is inherently collaborative, in that it involves many people coming together to examine a material assemblage of mutual interest by implementing a variety of tools and methods in sequence and in tandem. Independent projects establish organizational structures and information systems to help coordinate labour and pool information derived thereof into communal data streams, which can then be applied toward the production and publication of analytical findings. However, these information commons are not egalitarian; project communities establish practical expectations for how people in different positions should engage with archaeological information based on their roles. Contributing to and extracting from the commons is therefore scaffolded by diverse yet converging commitments to a collective enterprise, which are instilled through participation in a community of practice.

Through an abductive qualitative data analysis based on recorded observations, interviews, and documents collected from three cases, this paper highlights how digital systems designed to direct the flow of information do so via the coordination of labour and the strategic arrangement of human and object agency. Specifically, I highlight how the information systems that archaeologists rely on to generate records, internal reports, published papers and integrated datasets reify, reinforce, and sometimes challenge entrenched power structures and divisions of labour. For instance, I observe that each level of documentation effectively communicates local understanding to a broader collective by rendering prior work as series of formal processes performed by interchangeable agents; recognition of prior creative agency therefore diminishes as narratives about the objects of interest are further developed. At the same time, latent awareness of the practical circumstances, decisions and actions that contributed to records' creation enable them to be recontextualized, as needed. However, this ability to recontextualize data erodes at projects' boundaries, where familiarity with creative circumstances fades.

By characterizing the documentary media that archaeologists use to capture, organize and share knowledge as tools that facilitate controlled communication, and which govern how certain people may contribute to and access collectively-maintained knowledge, this paper re-casts the formation of data commons as a social rather than a technical enterprise. In highlighting some social and relational aspects of data work that are often overlooked when developing local and global research infrastructures, the paper puts forward a theoretical framework that explains the awkwardness of using other people's data, and the underwhelming results of initiatives to integrate data in contexts of reuse.

Introduction

- [1] Archaeological projects are inherently collaborative, in that they bring teams of people with a variety of unique outlooks and experiences to examine material assemblages of common interest. Archaeologists apply a multitude of tools and methods, in sequence and in tandem and across different kinds of work settings, to produce rich and heterogeneous data about their engagements with the archaeological record. They adhere to professional norms and expectations to organize the products of their collective labour, and rely on a combination of digital and analog devices and protocols to pool the outcomes of their respective tasks.
- [2] Advanced computational systems have addressed many technical barriers, but data integration is much more than just a technical process; it also involves considerable collaborative labour to facilitate commensurability of research outcomes and to ensure that all parties effectively contribute to common goals. In this paper I draw clear associations between project and data management, and their mutual reliance on technological systems as means of maintaining control. As such, it draws attention to the series of collaborative commitments and broader social contexts from which we develop data management systems and infrastructures.
- [3] Specifically, the paper examines how locally-circumscribed practices are re-framed as generic processes as part of attempts to streamline the dissemination of data, and how this permeates across project systems at various scales — albeit unevenly across various agents within project collectives.

Methods and Data

- [4] This paper articulates behaviours and attitudes of archaeologists who worked at two independent archaeological projects which serve as cases. [about the cases]
- [5] It is important to note that cases are not the subjects of inquiry, and instead represent discrete instances that share common reference to the overall research themes (Ragin 1992; Stake 2006). So while the cases are certainly not representative of the whole archaeological discipline, they do enable me to reveal some underappreciated systemic concerns that underlie data management. As such, the findings are informed by the behaviours and attitudes expressed to me by those who agreed to participate, and by my own standpoint as a scholar of the culture and practice of science and of the media and infrastructures that support data sharing, integration and reuse.
- [6] Through abductive qualitative analysis of observations of archaeological practice, interviews with archaeologists as they worked, and of documents they created, I captured diverse experiences and outlooks that provide insight regarding how data management infrastructures are valued and used in a variety of contexts.
- [7] The findings presented here are derived from my dissertation (Batist 2023), which produced three other papers deriving from the same methods and data. See Batist and Roe (2024), (**batist2025a?**), (**batist2025b?**) for further methodological details.

Findings

- [8] Analysis of archaeological materials and data is dependent on systems of control. To control the flow of materials and the structure of data, it is necessary to control people's behaviours. In other words, the epistemic value that a project produces is inherently dependent on the control that may be imposed on archaeologists' behaviours. Here I document some ways in which projects scaffold archaeologists' behaviours through technical and administrative means.

Sampling

- [9] In the cases I observed, specialist work often involved specific sampling procedures that differed from, but also occurred alongside, standard fieldwork practices. Specialists either collected their own samples or instructed fieldworkers to collect samples on their behalf and in ways that corresponded to their needs. Trench supervisors were usually present to guide the specialist in their trench, to help identify the locations from which specimens would likely yield more useful results, and to be aware of changes to their work environments that the sampling procedure may have caused. For example, Marie, who was Case A's OSL specialist, was responsible for establishing absolute dates for various stratigraphic units at the site, and was very involved in the careful work of collecting samples from the trenches.^[A73] OSL dating determines how long ago mineral grains were exposed to sunlight, and it is critical to avoid exposing specimens to light. Sampling therefore had to occur at night, and the involvement of trench supervisors was crucial for determining where to draw the sample under such cumbersome conditions.^[A96] Moreover, it was common practice to place a dosimeter where the sample was taken in order to record background conditions that would have affected that specific location, thus allowing calibration of the dose. Trench supervisors were instructed to take care not to disturb these dosimeters as they continued to excavate, until they would be collected at a later time.^[A97]
- [10] In some cases, as when a specialist could not be present during sample removal, it was deemed necessary to delegate sampling work to a fieldworker who received training in basic sampling procedures. This happened when a specialist had to leave the field before the end of an excavation season, and but had enough foresight to instruct a fieldworker to carry out the work on their behalf.^[A73] In this situation, the fieldworker effectively worked in service of a specialist activity and had to consider the criteria that would be most conducive to effective specialist analysis.
- [11] A similar phenomenon was observed at Case B, where Chris was considered an expert in collecting spatial coordinates using the differential global positioning system (DGPS) that forms the basis of their geospatial analyses. Trench supervisors called upon Chris to collect geospatial information from various targeted locations, and he depended on their understanding of the trench and of the value of specific spots that warranted data collection.^[B17] He also relied on trench assistants to help him by holding a portable DGPS "rover" (which communicates with a spatially-calibrated base station located in a fixed position), and talk among them tended to be about ensuring the proper collection of a reliable sample of points (see **?@fig-DGPS**).^[B18] As illustrated in Excerpt 1, Chris recorded the geospatial point at the position where Oliver, the trench assistant, placed the rover, while Liz, the trench supervisor, oversaw the process and identified what needed to be recorded.^[B18]

Liz: And then if you can just take one point there, and one point right in the middle, umm that's all we need.

Oliver: Ok.

Liz: And then you're free to start digging and I'll help you out in a minute.

. . .

Chris: So what are we doing?

Oliver: We're taking points.

Liz: Just one where you dug.

Oliver: Here?

Liz: Yep, yep. One in there, and then one over there is fine, somewhere in the middle.

. . .

Chris: It's excavation unit [redacted identifier]?

Liz: Yeah. Closing.

Chris: Alright.??

Excerpt 1: Communicating while collecting spatial data.

Finds processing

[12] Archaeological projects uncover a lot of material throughout the course of an excavation or survey, which must be sorted and processed to facilitate subsequent analysis. Finds processing begins in the field, where objects are first recovered and identified as finds that will yield valuable insight into the lives of past people. At both Cases A and B, fieldworkers divided these finds into broad-level categories corresponding to materials, such as ceramic, bone and shell, separating them into bags and buckets labelled with the excavation unit in which they were found, as they emerged from the trenches (see ?@fig-sieve: E and F) and ?@fig-sorting. At the same time, they tossed stones that exhibit no anthropogenic qualities (see ?@fig-sieve: D). They also used sieves to separate loose sediment from small objects, collecting those that they deem culturally meaningful, and discarded the rest (see ?@fig-sieve: A, B, C and D).

[13] Sometimes, fieldworkers set aside materials that they were unsure about and documented the findspot in more detail — “just in case” — before proceeding with their work.?? This documentation was tentative, and depending on whether a specialist would later determine that the find was worth saving for analysis, may not have become part of the official archaeological record.??, ??

[14] It is noteworthy that fieldworkers were expected to classify finds on the basis of characteristics commonly recognized as more natural, while study of their anthropogenic qualities was typically reserved for finds analysis.?? In effect, a distinction was made between fieldwork, which deals with *materials*, and finds analysis, which deals with *artefacts* and *ecofacts*. Moreover, even when specialists worked in the field, they operated as fieldworkers; their actions were constrained by the practical circumstances of fieldwork and the position of fieldwork within a broader apparatus of archaeological knowledge production. In this sense, the division of labour was partly due to pragmatic concerns, namely the fact that fieldworkers were usually occupied with their own specific tasks that required mental concentration, especially when instructed to do this work quickly.

[15] After hauling all the finds back to the dig house, they needed to be cleaned by brushing the finds with water.??, ?? Trench assistants performed this task after they have returned from the field, remaining productive even during the late afternoons when it was too hot to perform strenuous outdoor labour.?? In my

unrecorded observations I noticed that people talked about the day's work and their personal and collective experiences, and speculated about the things they recovered and which they were then revisiting in a secondary context. As non-experts, their amateur analyses remain unexplored, but they still felt an impetus to examine these objects and try to develop insights from these experiences.

[16] Cleaned materials were then left out in the sun to dry before being brought to a space dedicated for finds analysis and storage.^{??} The materials were dumped on a broad table, and specialist's assistants quickly sorted through them. They discarded any non-anthropogenic materials collected in error, then counted and weighed all remaining artefacts, before performing further analysis based on their typological and technological characteristics.^{??}

[17] At Case A, I observed how Jolene, a lithics specialist, went through and sorted the material on the table, sometimes mumbling her reasoning aloud.^{??} After she settled on her arrangement, she dictated her description of the assemblage as a whole, in a fragmented tone that insinuates strong intention behind each word, and Maude, her assistant, wrote down this speech.^{??} Maude then recorded the quantities of artefacts in each category, as laid out by Jolene, and entered this information into a spreadsheet.^{??} Whereas Jolene was the source of creative knowledge production, Maude was responsible for embedding these insights within the project's organizational apparatus.^{??}

[18] Maude also maintained a checklist to track all aspects of work performed on the materials, including information about the materials' provenance, who processed the materials, when the work occurred, and detailed parameters that structured processing tasks.^{??} This checklist resembled the records maintained by the project's database manager to trace what records await integration into the database and to ensure that all information was properly recorded.^{??, ??}. These records existed to track progress and verify the integrity of work. No aspect of finds processing involved maintaining a journal or recording substantial interpretations of the materials, which revealed a failure to recognize that any meaningful insights could arise during these stages of work. Finds processing was therefore positioned as a means to an end, that being the arrival of cleaned and sorted finds upon the specialists' desks, where meaningful archaeological knowledge would be produced.^{??, ??, ??, ??}

Integrating and aligning perspectives

[19] After performing their work, specialists reported back to the project with their findings. This involved providing the project with documents that contained and explained the analytical results. These documents typically comprised electronic spreadsheets and typeset documents, but sometimes also included lab journals, paper recording sheets, source code, and digital images. These media varied in their functional roles and value.

[20] For instance, the spreadsheets that archaeological specialists produce in the cases I examined were the formally organized results of systematic analysis. They were organized on a sample-by-sample basis, associated with discrete units from which each sample element was taken (see **?@fig-archaeometry-spreadsheets**).^{[A6], [A98]} These records were analytical results in that they were the products of analytical processes, but they were also data in the sense that they were considered to be grounded and stable records of systematic observation that may serve as a sound basis for further investigation.

- [21] Specialist reports, on the other hand, were summaries of a specialist’s findings. They described overall trends that specialists observed in their preliminary analysis of the findings, highlighted their potential significance, and proposed ways to ameliorate the results or relate them to other aspects of the project.^[B11] In other words, specialist reports contextualized the analysis and related the findings to the project’s general research objectives. They presented conclusions that aligned or conflicted with alternative perspectives, settled or provided greater confidence to particular claims, and suggested aspects of work that warranted further attention. They did these things in an embodied manner, and in ways that reflected the pragmatic situation of their work; they served to contextualize the data presented in spreadsheets, which lacked any such capability on their own.
- [22] Specialist datasets were integrated into projects’ relational databases by importing the relevant records as their own sets of tables, with explicit links drawn between them and the database backbone, which reflected the organizational structure of archaeological entities encountered in the field (see **?@fig-archaeometry-spreadsheets**).^[A98] For example, samples that were drawn from a particular excavation unit were typically listed in a specialist table, but were linked to an overarching index of all excavation units.^[A6] Analytical results could therefore be retrieved for particular parts of a site, alongside findings derived from different kinds of analysis performed on material deriving from the same contexts.
- [23] The material that was physically distributed and sent out for specialized analysis was thus conceptually re-integrated in informational terms. Diverse kinds of material were rarely examined in close physical proximity outside of fieldwork, though the potential to integrate abstract measurements about them was highly valued.^{[B19], [C17], [C18], [C19]} However, this actually rarely happened in ways that leveraged the computational potential of these efforts. Specialists provided their own preliminary analyses in written and oral reports, in which they examined the materials under their own purview.^{[B1], [B11]} Analysis was therefore siloed, and findings deriving from these analyses were subsequently integrated.^{[A99], [A89], [B12]}
- [24] While specialists’ interests may have overlapped with those of other members of the project, directors preferred to have one specialist for each means of investigation. At the same time, some degree of overlap among specialists who examine the same kinds of materials from different methodological perspectives was expected and encouraged. For example, the geoarchaeologists at Case A shared a strong affinity based on their mutual work with geological materials, despite the fact that they specialized in different geological processes.^{[A85], [A95]} This benefited the project immensely, since it allowed each specialist to reflect upon the limitations of their approach and to harmonize that approach with those of the other specialists.
- [25] At the same time, however, the value of these complementary perspectives has yet to be demonstrated when presenting findings to the research community. Much of the discussion among geoarchaeologists at Case A, for instance, concerned strategic planning, including planning where to excavate, prioritizing where to take samples, and considering the degree of confidence that should be placed in certain findings. In other words, these were process-oriented discussions that reflected on decision-making processes. This information was not effectively represented in the database and in reports, which were more meant to articulate research outputs in a clear and confident manner. Inputting analytical findings into a database was therefore a lossy process that failed to capture much of the pragmatic knowledge and critical reflection provided by individuals and collectives.

Informal consultation

[26] Additional interactions between archaeologists and external members of their research communities also occurred through consultation with key figures, whom projects called in to either provide informed insights as a prelude to broader dissemination of a project's findings or to provide expert opinions on how work should proceed. Consultants were specialists in their own right, but their involvement was usually limited to one-off interactions that precluded long-term commitment to the project.

[27] As Case A, a consultant was called in to examine a peculiar and anomalous material configuration that confounded project participants. As illustrated in Excerpt 2, Basil asked Andreas, a micromorphology specialist, to perform a preliminary, *ad-hoc* examination of the microstratigraphy in a trench that exhibited evidence of hearth activity:[A118], [A119], [A120], [A121]

Andreas: And on this side, on this side, is the, the fire base, sand, sand right below the, the hearth. ...This sand is de-calcified, which again implies kind of, [left on] the surface in order to de-calcify. And on this surface, the guys came and put fire. Actually the matrix of the black stuff is sand again. ...So the whole thing is kind of in play but not in sequence. ...Of course we don't see any more ashes because they have been dissolved away. Actually what you see under the microscope is just sand grains surrounded by [unclear] made of very fine charcoaled dust, dust charcoal. That's why you don't find, I guess, charcoal inside, probably [a] few bits.

Basil: Yeah, it's nothing. Yeah it's a very very tiny amount.

Andreas: Yeah, I mean okay, it's like putting a fire down on the sand. You know, the sand is something that moves, always, around. Even people when we're trampling it, you know animals, you know Sadly, nothing would survive, ...so much not in a ...but would end up in dust, something like that.

Alfred: Oh well.

Andreas: But still, it's [held up] in place, and made on the, on the material that is just below. So they are, let's say, kind of the same, umm, story.

Basil: Excellent.

Alfred: That's good. Yeah.

Andreas: I mean you can date, ...The accumulation of the sand below and the fireplaces on top, they might be about the same age, more or less. I mean, if you are lucky and you ...you can separate these two events, it would be nice but I think it's, it wouldn't be very...

Alfred: [unclear, but something about carbonized remains]

Basil: We have a few carbonized remains. And we, we just took another 167 litres of flotation so our, we have our archaeobotanist coming through next week.

Andreas: Yeah, then of course. ...It is, there's no doubt about it.

Alfred: Yeah.

Andreas: Actually yeah, ok I forgot. There's quite a few burnt bone inside but are reduced in the size of sand. Let's say less than a half a millimetre, but they are burnt. Again, because you know, they are so [can't hear because of the sieve nearby], they've degraded in that small sizes.

Basil: Well, that's the only bone we have from the site so far.

Andreas: I can show you under the mic, I mean they are tiny like ...sand grains. And they are burned too.

Basil: I wonder if we could do anything further with that.

Andreas: Other than dating, I don't think, what else? Yeah. Yeah, I think more or less, you can't do more.

Basil: So what—

Andreas: Ahh, I mean, ok, in theory, some more fancy biochemical analysis? I don't know, I don't know if these things have survived. But why not? There have been, lately, some umm biochemical analyses on these kinds of stuff.??

Excerpt 2: Andreas provides guidance on how to collect meaningful information.

[28] In this exchange, Andreas remarked that there was enormous potential to apply more in-depth micromorphological and instrumental analyses in this particular trench. He provided guidance on how to expand excavation around this trench on the basis of his understanding of the geological processes at work on the site and how they would affect the archaeological traces of hearths, and he suggested new forms of analysis that the director had not yet considered. Each participant played a particular role here: Andreas provided his brief analysis as a visiting and non-committed specialist; Alfred, who was the project's dedicated micromorphology specialist, recalled specific aspects that can be leveraged to support his own investigations; and Basil used his authority as project director to decide that these ideas were worth pursuing in greater depth.^{[A120], [A121]}

[29] Similarly, since Case B did not employ a full-time zooarchaeologist, Chris (who was also one of the co-directors) occasionally sent pictures of faunal remains they recovered to a colleague who would quickly identify them. For example, I observed one instance when Chris was fairly confident that a long bone they found was a cow's femur but he wanted to verify that it was not a human bone, which would have called for very different excavation procedures and project management decisions.^[B31]

[30] Additionally, I observed that projects sometimes seek to leverage consultants' authority on a subject to bolster confidence in cases when their findings rested on shaky foundations.^{[A122],[A123][A124]} This was exemplified by the way that Case A handled its anxieties regarding the extremely colluvial nature of the site. Colluviation complicates any effort to date stratigraphic units and had resulted in many artefacts having extremely weathered surfaces that make them more difficult to characterize. Despite his confidence in their claims due to his daily involvement examining these artefacts, and his acknowledgement that the stratigraphy is imperfect, Basil knew that he would have a hard time convincing others of the nature of the things that they found. He therefore involved Denise, a major figure in lithics analysis who was responsible for publishing the most definitive assemblages of the kind of materials recovered at Case A, to examine their materials, as articulated in Excerpt 3:^{[A124],[A125]}

Basil: ... for decades, there have been accepted means of scientifically representing the archaeological record, umm to act as evidential bases for your claims. It's understood, that you know, it's impractical for every single interested archaeologist to come to see your site, to see your material. You could be in Australia and the site could be in Peru, umm and so, without that tactile, immediate relationship with the evidence basis that somebody's uhh excavated, there are ways of representing it through photography and line illustration. Umm, that always used to seem to be enough, but it seems like again, it comes down to this contentious nature of our site. The fact that we are, we allegedly have such an early site where such early sites are not meant to be, I'm starting to discover that some of what used to be the accepted means of, accepted ehh evidential umm bases, don't seem to be good enough for everyone. And so it's been very important to have people like Denise come and see this stuff in the flesh. Yes, she's seen the drawing, yes, she's seen the photograph, but for her to see it first hand, handle it, pick it up, query it, umm look at, look at assemblages, as opposed to cherry picked illustrated pieces, has been fundamental to convince her that we really have what we say we have.^[A124]

Excerpt 3: Basil explains his rationale for inviting Denise to examine the archaeological material.

[31] At the same time, Denise informed Basil about some of her concerns with the material, and in Excerpt 4 Basil recalls how this constructive criticism provided him with an understanding of how to proceed to accommodate these issues.^{[A125],[A126]}

Basil: So the superstar material that Lauren was excavating, that we thought was Levallois blade cores, she disagreed. She thinks the material is fabulous, she's never seen this material before in Greece, she thinks it's Aurignation and looks much more like what you would find in France. She's blown away. She's blown away by—

Zack: But it's still Middle Pal?

Basil: Very early Middle Pal. But, we had shown her our superstar pieces. This is where it—

Zack: I was in the trench that first day when we, and I was like cores, wow, tons of them.

Basil: I mean that stuff is— but we had shown her some of our superstar pieces from the survey, like here's a Levallois point, classic thing. But she's like hmm, it's just one piece. Haha you know, it's just like, she's not exactly [unclear], but she has very high standards.

Zack: So is she sticking with, is she saying that Middle Pal was a thing though?

Basil: She was like, so she was looking at these, these little bits and pieces, and like, well, the stuff, you know, we showed her a whole bunch of stuff, it's like, we think this is Levallois, she's like no, it's this. And then we show her some stuff from the survey and it's like, hmm it's one piece and it could be, but it's just one piece, that's not really enough. And then finally we showed her the material, Jolene showed her material from [the trench], Alice's new trench, and she's like yes, this is Middle Palaeolithic.??

Excerpt 4: Basil explains the feedback he received from Denise, and what this means for future work.

[32] From these examples we see that members of the archaeological community who were not active project participants sometimes supplemented the input provided by specialist project participants. As people who were external to the project, they were not committed to following through on their advice; they instead relayed their interpretations to specialists who are already members of the project and who were capable of implementing their guidance.^{[A127],[A128]} Moreover, as outsiders, they were not forced to adhere to the systematic recording systems that scaffolded internal procedures, and the informal nature of their contributions was in fact a crucial aspect that imbued them with value.

Discussion and Conclusion

[33] Projects experience immense pressures to render observations in ways that are amenable to formal and functional analysis, which in turn mandate rigid and decisive data collection processes to instill confidence in the stability and evidentiary nature of archaeological data.^{??, ??} As such, particular kind of analytic process comes to dictate the conduct of other aspects of work, and involves reducing the interpretive agency of actors involved in all aspects of the data's constitution and processing.

[34] This results in prioritization of archaeological materials as formally structured and abstract entities, devoid of character in their own right and always perceived as specific instances of a more general class. The only person who can recognize a thing as meaningful is someone who is deemed capable of granting it meaning. This is exemplified by my observations of work performed by trench assistants, who essentially serve as unskilled labourers who are shuffled around a project to perform menial tasks that, almost exclusively, concern materials that have not yet been assigned archaeological character (**batist2025?**). Beyond operating as sensory instruments, fieldworkers serve others' creative efforts by cleaning artefacts and performing preliminary sorting tasks. Those involved in this type of work prepare materials for analysis that others will perform, namely specialists. Similarly, trench assistants move sediment and rocks out of a trench, but it is up to the supervisor, who is responsible for recording the significance of these materials, to make meaning out of them. Trench assistants are cast as mere sensors, whose identifications of a thing, if taken seriously at all, are based on the character of shapes and textures they encounter, and not on any underlying archaeological theory or

model.

[35] Their interchangeability, owing to the limited expectations made of them (simply responding to prompts), renders fieldworkers invisible and removes any potential for imagining data collection as discursive and situated. This supports the absorption of fieldworkers into distributed data-recording system, thereby reducing the impacts of their situated perspectives and presenting their work as a form of objective sensory understanding.

[36] At the same time, interpretation is delegated to others who are not as present at the site, and whose distance from the archaeological encounter provides a kind of freedom to take the data at face value, rather than to grapple with their discursive origins. This contributes to a sense that so-called “data-driven” analysis generates new knowledge synthesized from a series of objective facts derived from generic, predictable, and protocol-driven sensing devices (Huggett 2022). Archaeological information infrastructures are thus designed to enable the appropriation of direct observations by those with the means of extracting more meaningful value from them.

[37] Moreover, the paper observed how there is greater flexibility for those higher up in the chain to explain the context of the work they perform. However, the advent of open data sharing, integration and reuse adds another layer of abstraction, whereby entire projects and the totality of the data they collect are rendered as generic instances of some greater class. The discomfort that project directors sense upon having their creative agency reduced is akin to that which was experienced during the advent of tighter control over fieldwork recording systems (as documented by Chadwick (1998) and Thorpe (2012)), and which we now take for granted.

[38] This paper frames the technologies that archaeologists use to re-situate individual archaeologists in relation to project collectives as means through which managerial protocols are carried out. Specifically, it documents a few underappreciated aspects of archaeological projects’ sociotechnical arrangements that should be accounted for more thoroughly:

1. Systems designed to direct the flow of information do so via the coordination of labour, and the strategic arrangement of human and object agency.
2. These systems are *attempts* at control, which are actively resisted by those who work under them, and passively resisted by the wild nature of the things that they attempt to tame.
3. Informal and socially-mediated forms of scholarly communication are de-prioritized and excluded from data streams ascribed with greater scholarly legitimacy, despite the fact that this information is crucial for developing a sense of mutual understanding that underpins comprehension of formal datasets.
4. Boundaries, whether they restrict membership within a collective, delimit a project’s scope, or limit the time frame under which a project operates, have practical positive value, and are not just arbitrary impediments.
5. Information systems and the institutional structures that support them tend to reinforce and reify

existing power structures and divisions of labour, including implicit rules that govern ownership and control over research materials and that designate who may benefit from their use.

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Competing Interests

- [41] The author states no conflicts of interest.

Author Contributions

- [42] Zachary Batist is the sole author of this work. He defined the scope of the study and identified suitable cases for inclusion, collected and processed all data, performed analysis, interpreted the findings, created all the figures, and wrote the paper.

Informed Consent

- [43] Informed consent has been obtained from all individuals included in this study, in compliance with the University of Toronto’s Social Sciences, Humanities, and Education Research Ethics Board, Protocol 34526.

Data Availability Statement

- [44] The data generated and analyzed during the current study are included in this published article’s supplementary files.

Author Bios

- [45] Zachary Batist obtained his PhD from the University of Toronto’s Faculty of Information. His research explores the collaborative commitments inherent throughout archaeological practice, especially relating to data management and the constitution of information commons. He currently works as a Postdoctoral Researcher at the Department of Epidemiology, Biostatistics and Occupational Health in the School of Public and Global Health at McGill University, where he investigates the collaborative, technical and administrative structures that scaffold data harmonization in epidemiological research.

Figure Captions

Supplementary Materials